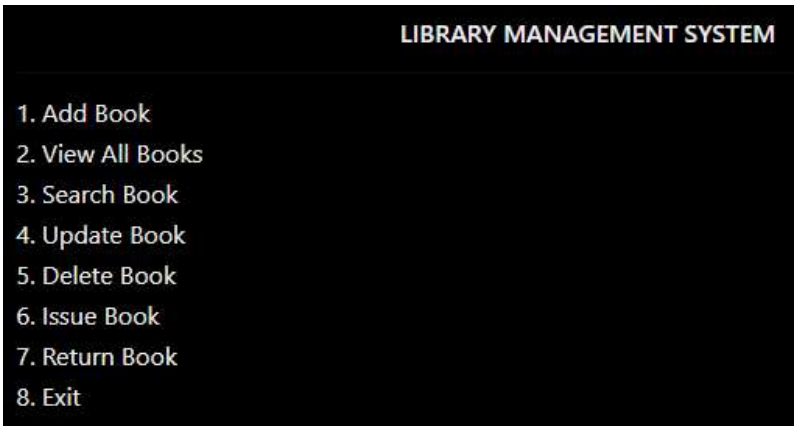
Assignment # 04: Library Management System

Objective:

- Create a menu-driven Python program that allows a librarian to manage books in a library.
- Develop logical thinking.
- The program should allow the user to add books, view all books, search for a book, update book details, delete books, issue books, and return books.

Instructions:

1. When the program starts, it should display a menu similar to this:



   The user should select an option by entering a number. After completing that operation, the program should display the menu again until the user chooses Exit.

2. Each book should have the following information:

Field	Example
Book ID	101
Title	Python Basics
Author	John Smith
Total stock	5
Available copies	3

   Use a dictionary to represent one book and a list to store multiple books.

Feature 1 — Add Book

1. Ask for the book ID, title, author, and number of copies.
2. Create a dictionary for the new book.
3. Add it to the `books` list.
4. Display a success message.

Challenge: Do not allow duplicate book IDs.

Feature 2 — View All Books

1. Check whether the list contains any books.
2. If empty, display “No books available.”
3. Otherwise, use a `for` loop to display every book's details.

Feature 3 — Search Book

1. Ask the user for a book ID.
2. Search through the list using a `for` loop.
3. If the ID matches, display the book's details.
4. Otherwise, display “Book not found.”

Feature 4 — Update Book

1. Ask for the ID of the book to update.
2. Find the matching book.
3. Ask for the new title, author, or copy count.
4. Update the selected dictionary.
5. Display a confirmation message.

Feature 5 — Delete Book

1. Ask for the book ID.
2. Find the book in the list.
3. Remove the matching record.
4. If it does not exist, display an appropriate message.

Feature 6 — Issue Book

1. Ask for the book ID.
2. Find the book.
3. Check whether copies are available.
4. If available, decrease the copy count by one.
5. If no copies remain, display “Book is not available.”

Feature 7 — Return Book

1. Ask for the book ID.
2. Find the book.
3. Increase its available copy count by one.
4. Display a success message.

Challenge: Prevent returns from increasing stock beyond the number of copies the library owns. For accurate tracking, record issued books or loans.